



UNIVERSITY OF PETROLEUM AND ENERGY
STUDIES
DEHRADUN

Applied DevOps

Experiment 7

MTECH - School of Computer Science
CYBER SECURITY AND FORENSICS

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Objective — To develop and deploy a web application using containerization with Docker, automate infrastructure setup using a Dockerfile, and validate deployment consistency by running multiple container instances from the same image.

Task to be Performed

Task 1: Set Up Working Environment

```
docker-exp7 — -zsh — 124x43

Last login: Tue Mar 31 16:48:06 on ttys000
[sheoraninfosec@Jigeshs-MacBook-Air ~ % pwd
/Users/sheoraninfosec
[sheoraninfosec@Jigeshs-MacBook-Air ~ % mkdir docker-exp7
[sheoraninfosec@Jigeshs-MacBook-Air ~ % ls
Applications
cd-demo-app
ci-calculator
Creative Cloud Files 500087265@stu.upes.ac.in 0efd2be8340bd3b1f8aec782ef0cc2099fd5df328bf375f115a125577e95d625
Creative Cloud Files Personal Account jigeshsheoran2002@gmail.com 4EB975E15F4FED470A495EAE@AdobeID
Desktop
devops-lab-demo
docker-exp7
Documents
Downloads
dvcs-lab
dvcs-lab-clone
iCloud Drive (Archive)
lambda-exp
Library
misp-docker
Movies
Music
ns2-docker
Pictures
Public
Virtual Machines.localized
[sheoraninfosec@Jigeshs-MacBook-Air ~ % cd docker-exp7
sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 %
```

Task 2: Develop Simple Web Application

Step 1: Create HTML File

```
docker-exp7 — nano index.html — 124x43

UW PICO 5.09 File: index.html

<!DOCTYPE html>
<html>
<head>
  <title>Docker Experiment 7</title>
</head>
<body>
  <h1>Infrastructure Automation using Docker</h1>
  <p>This web page demonstrates deployment consistency using containerization.</p>
</body>
</html>
```

Task 3: Create Dockerfile

Step 1: Create Dockerfile

```
docker-exp7 — nano Dockerfile — 124x43
File: Dockerfile
UW PICO 5.09

FROM nginx:latest

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80
```

FROM nginx → base image

COPY → injects your app into container

EXPOSE → tells Docker which port to use

Task 4: Build Docker Image & Run Container

```
docker-exp7 — -zsh — 124x45

sheorani@Jigeshs-MacBook-Air:~$ docker build -t docker-exp7 .
zsh: command not found: docker
sheorani@Jigeshs-MacBook-Air:~$ docker --version
Docker version 29.3.1, build c2be9cc
sheorani@Jigeshs-MacBook-Air:~$ docker build -t docker-exp7 .
[+] Building 11.6s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 116B
=> [internal] load metadata for docker.io/library/nginx:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 277B
=> [1/2] FROM docker.io/library/nginx:latest@sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
=> => resolve docker.io/library/nginx:latest@sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
=> => sha256:881ff0f5a79a3151be8736c639f93b0a92dd46e4f47d3487d98d2099a5652e5 1.21kB / 1.21kB
=> => sha256:5716fd8931326782b7a67170c921932c06b391876f0f11eb11fe102e351dce61 1.39kB / 1.39kB
=> => sha256:0d9786b44f2a72c3a7e982ea1a5e90172039475493cf05dea585480950b09ce0 400B / 400B
=> => sha256:e14d31524fc89f789831d4539ccbb3780cb216e786d96cc30c0b6572befdcb48 953B / 953B
=> => sha256:0a43e039475a9aad5a5e415b4551a1576b21b8f272f26163cf92bf235d61c131 626B / 626B
=> => sha256:b1a5e9a47bf9c1783a0585f735d73e76cddf97a26a8f468b7f8758fe1e927a06 31.14MB / 31.14MB
=> => sha256:f4badebdec24858ef2dc51256f6418328e305e9c3c5a5e093581f83425618bd5 30.14MB / 30.14MB
=> => extracting sha256:f4badebdec24858ef2dc51256f6418328e305e9c3c5a5e093581f83425618bd5
=> => extracting sha256:b1a5e9a47bf9c1783a0585f735d73e76cddf97a26a8f468b7f8758fe1e927a06
=> => extracting sha256:0a43e039475a9aad5a5e415b4551a1576b21b8f272f26163cf92bf235d61c131
=> => extracting sha256:e14d31524fc89f789831d4539ccbb3780cb216e786d96cc30c0b6572befdcb48
=> => extracting sha256:0d9786b44f2a72c3a7e982ea1a5e90172039475493cf05dea585480950b09ce0
=> => extracting sha256:881ff0f5a79a3151be8736c639f93b0a92dd46e4f47d3487d98d2099a5652e5
=> => extracting sha256:5716fd8931326782b7a67170c921932c06b391876f0f11eb11fe102e351dce61
=> [2/2] COPY index.html /usr/share/nginx/html/index.html
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:e572c055d2633567057c54b89ba108fa6f01bf2bc727a6cb3a19bc43fd68bd07
=> => exporting config sha256:77b9678cde0d34e004faf5688b7c6d85b7d5563038add01fba26309fbc94e7a
=> => exporting attestation manifest sha256:3096b65b6b3d55b7cd1f9f799c61c8b4abaa16b6b026029359850fcfadcd282
=> => exporting manifest list sha256:f76ba752c839d1e995832d86188c22e3d937419ee9f1451a38d73b2e05829d9f
=> => naming to docker.io/library/docker-exp7:latest
=> => unpacking to docker.io/library/docker-exp7:latest
sheorani@Jigeshs-MacBook-Air:~$ docker images

```

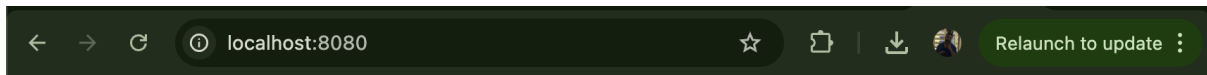
IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
docker-exp7:latest	f76ba752c839	255MB	61.3MB	

```
sheorani@Jigeshs-MacBook-Air:~$ docker run -d -p 8080:80 docker-exp7
7ce0bacd5227a48f8839a6b32f3442755f092ad0ce0fb4a83b363d8d326ea38c
sheorani@Jigeshs-MacBook-Air:~$
```

-d → run in background

-p 8080:80 → map host → container

When you open your browser , you will see you deployed webpage at **http://localhost:8080**



Infrastructure Automation using Docker

This web page demonstrates deployment consistency using containerization.

Task 6: Validate Deployment Consistency

Step 1: Stop Container

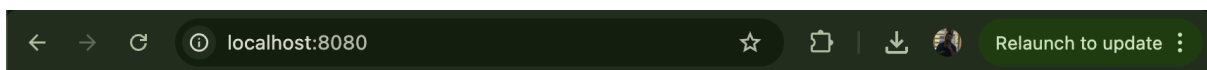
```
[sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 % docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
7ce0bacd5227   docker-exp7  "/docker-entrypoint..." 3 minutes ago  Up 3 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
wonderful_cray
[sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 % docker stop 7ce0bacd5227
7ce0bacd5227
[sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 % ]
```

Step 2: Remove Container

```
docker-exp7 — -zsh — 124x45
[sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 % docker rm 7ce0bacd5227
7ce0bacd5227
[sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 % ]
```

Step 3: Run Again

```
docker-exp7 — -zsh — 124x45
[sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 % docker run -d -p 8080:80 docker-exp7
17fd3446986bf9430a2bf40398f7f7e8052a13f44436ebc97606ea69b9a3ff40
[sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 % ]
```



Infrastructure Automation using Docker

This web page demonstrates deployment consistency using containerization.

Task 7: Push to Docker Hub

Step 1: Login

```
docker-exp7 — -zsh — 124x45

[sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 % docker login

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: ZCTB-THDX
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

Login Succeeded
sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 %
```

Step 2: Tag image and push to DockerHub

```
docker-exp7 — -zsh — 124x45

[sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 % docker login

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: ZCTB-THDX
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

Login Succeeded
[sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 % docker tag docker-exp7 dwc001/docker-exp7
[sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 % docker push dwc001/docker-exp7
Using default tag: latest
The push refers to repository [docker.io/dwc001/docker-exp7]
f4badebdec24: Pushed
b9943749570d: Pushed
a8b3db0149ea: Pushed
0d9786b44f2a: Pushed
881ff00f5a79: Pushed
b1a5e9a47bf9: Pushed
0a43e039475a: Pushed
e14d31524fc8: Pushed
5716fd893132: Pushed
latest: digest: sha256:f76ba752c839d1e995832d86188c22e3d937419ee9f1451a38d73b2e05829d9f size: 856
sheoraninfosec@Jigeshs-MacBook-Air docker-exp7 %
```

Observation

- Image successfully uploaded to public registry.
- Can be reused anywhere.

Task 8: Questions for Analysis

1. How Docker enables infrastructure automation

- Docker uses Dockerfile to define environment setup.
- Eliminates manual installation/configuration.
- Ensures repeatable deployments.
- Reduces human errors.

2. How containerization ensures consistency

- Same image runs everywhere.
- No dependency conflicts.
- Environment remains identical.
- Eliminates "works on my machine" problem.

3. Docker Image vs Container

Feature	Image	Container
Definition	Blueprint	Running instance
State	Static	Dynamic
Example	nginx image	Running nginx container

4. Advantages over traditional setup

- Faster deployment
- Lightweight (no full OS)
- Easy scaling
- Isolation (important for security)
- Portability across systems

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